

Ian Waudby-Smith

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Academic positions

University of California, Berkeley
Miller Postdoctoral Fellow in Statistics
Host: Michael I. Jordan

Berkeley, CA
2024–2027

Education

Carnegie Mellon University
PhD, Statistics
Advisor: Aaditya Ramdas
Umesh K. Gavaskar Best Dissertation Award

Pittsburgh, PA
2019–2024

University of Waterloo
BMath, Pure Mathematics & Statistics
5-year co-op program
Dean's Honours List

Waterloo, Ontario
2013–2018

Papers

($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith***. Gaffke's confidence interval for the mean of bounded data is inadmissible but asymptotically efficient. *arXiv preprint arXiv:2607.18661*, 2026.

($\alpha\beta$) Jiahao Ming, Aaditya Ramdas, Yi Shen, Ruodu Wang, and **Ian Waudby-Smith***. Combining e-values via demi-supermartingales. *arXiv preprint arXiv:2603.10329*, 2026.

Ricardo J. Sandoval*, **Ian Waudby-Smith***, and Michael I. Jordan. Multi-armed sequential hypothesis testing by betting. *arXiv preprint arXiv:2603.17925*, 2026.

Ben Chugg*, Etienne Gauthier*, Michael I. Jordan, Aaditya Ramdas, and **Ian Waudby-Smith**. Post-hoc large sample statistical inference. *arXiv preprint*, 2026.

Ricardo J. Sandoval, Sivaraman Balakrishnan, Avi Feller, Michael I. Jordan, and **Ian Waudby-Smith**. On nonasymptotic confidence intervals for treatment effects in randomized experiments. *arXiv preprint*, 2026.

Ian Waudby-Smith, Ricardo J. Sandoval, and Michael I. Jordan. Universal log-optimality for general classes of e-processes and sequential hypothesis tests. *arXiv preprint*, 2025+.

Ian Waudby-Smith, Martin Larsson, and Aaditya Ramdas. Nonasymptotic and distribution-uniform Komlós-Major-Tusnády approximation. *arXiv preprint*, 2025+.

Ian Waudby-Smith, Edward H. Kennedy, and Aaditya Ramdas. Distribution-uniform anytime-valid sequential inference. *arXiv preprint*, 2023+.

($\alpha\beta$) Johannes Ruf and **Ian Waudby-Smith**. Concentration inequalities for strong laws and laws of the iterated logarithm. *Bulletin of the London Mathematical Society (accepted)*, 2026.

Ian Waudby-Smith, Martin Larsson, and Aaditya Ramdas. Distribution-uniform strong laws of large numbers. *The Annals of Applied Probability (accepted)*, 2026.

Ian Waudby-Smith, David Arbour, Ritwik Sinha, Edward H. Kennedy, and Aaditya Ramdas. Time-uniform central limit theory and asymptotic confidence sequences. *The Annals of Statistics*, 52(6):2613–2640, 2024.

Ian Waudby-Smith, Lili Wu, Aaditya Ramdas, Nikos Karampatziakis, and Paul Mineiro. Anytime-valid off-policy inference for contextual bandits. *ACM/JMS Journal of Data Science (Invited contribution)*, 1(3):1–42, 2024.

Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B: Statistical Methodology (Discussion paper)*, 86(1):1–27, 2024.

Ian Waudby-Smith, Zhiwei Steven Wu, and Aaditya Ramdas. Extensions of randomized response for private confidence sets. *International Conference on Machine Learning (Oral presentation)*, 2023.

Akash V. Maharaj, Ritwik Sinha, David Arbour, **Ian Waudby-Smith**, Simon Z. Liu, Moumita Sinha, Raghavendra Addanki, Aaditya Ramdas, Manas Garg, and Viswanathan Swaminathan. Anytime-valid confidence sequences in an enterprise A/B testing platform. *The ACM World Wide Web Conference*, 2024.

Ian Waudby-Smith, Philip B. Stark, and Aaditya Ramdas. RiLACS: Risk limiting audits via confidence sequences. In *International Joint Conference on Electronic Voting (Best paper award)*, pages 124–139. Springer, 2021.

Ian Waudby-Smith and Aaditya Ramdas. Confidence sequences for sampling without replacement. *Advances in Neural Information Processing Systems (Spotlight)*, 33:20204–20214, 2020.

Ian Waudby-Smith, A Simon Pickard, Feng Xie, and Eleanor M. Pullenayegum. Using both time tradeoff and discrete choice experiments in valuing the EQ-5D: Impact of model misspecification on value sets. *Medical Decision Making*, 2020.

Ian Waudby-Smith, Nam Tran, Joel A. Dubin, and Joon Lee. Sentiment in nursing notes as an indicator of out-of-hospital mortality in intensive care patients. *PloS one*, 13(6), 2018.

($\alpha\beta$) Alphabetical ordering.

*Equal contribution.

Teaching Experience

Carnegie Mellon University

Graduate Teaching Assistant

- 36-708: Statistical Methods in Machine Learning (x2)
- 36-462: Data Mining
- 36-401: Modern Regression
- 36-731: Foundations of Causal Inference
- 36-732: Modern Causal Inference
- 10-880: Game-theoretic Probability, Statistics, and Learning

Pittsburgh, PA

2019–22

Other experience

Google Research

Student Researcher

Mentors: Jean Pouget-Abadie & Jennifer Brennan

New York, NY

Jun–Aug 2023

Microsoft Research

Research Intern

Mentor: Paul Mineiro

New York, NY & Redmond, WA

May–Aug 2022

Adobe Research

Research Intern

Mentors: David Arbour & Ritwik Sinha

San Jose, CA

Jun–Aug 2020

The Hospital for Sick Children (SickKids)

Research Student

Mentor: Eleanor Pullenayegum

Toronto, Ontario

Apr–Aug 2019

Health Data Science Lab, University of Waterloo

Research Assistant

Mentors: Joel Dubin & Joon Lee

Waterloo, Ontario

2016–18

Department of Statistics, University of Waterloo

Research Assistant

Mentor: Pengfei Li

Waterloo, Ontario

Apr–Aug 2017

Department of Pure Mathematics, University of Waterloo

Research Assistant

Mentor: Yu-Ru Liu

Waterloo, Ontario

Jan - Apr 2017

Cancer Care Ontario

Student Analyst

Toronto, Ontario

Jan–Apr 2016

Mentor: Zhihui (Amy) Liu

Service

Thesis committee member: Tyron Lardy (PhD in Statistics, Leiden University), Noah Liniger (MSc in Data Science, ETH Zurich)

Reviewer: The Annals of Statistics, Journal of the Royal Statistical Society (Series B), Bernoulli, Biometrika, The Electronic Journal of Probability, The Electronic Journal of Statistics, The Journal of the American Statistical Association, STOC, The New England Journal of Data Science, Sankhya A.

University of California, Berkeley

Berkeley, CA

Volunteer

- Miller symposium organizing committee
- Incoming Miller fellow initiative

Carnegie Mellon University

Pittsburgh, PA

Volunteer

- Organizer of the Statistical Machine Learning Reading Group (SMLRG)
- Women in Data Science (WiDS) conference volunteer
- Computing committee student representative
- Incoming PhD student mentor

Awards

Institute of Mathematical Statistics

Berkeley, CA

New Researcher Travel Award

2026

Miller Institute for Basic Research in Science

Berkeley, CA

Miller fellowship

2024–'27

Carnegie Mellon University Department of Statistics and Data Science

Pittsburgh, PA

Umesh K. Gavaskar Best Dissertation Award

2024

Statistical Society of Canada

St. John's, NL

Probability Section Student Research Presentation Award

2024

Amazon Science

Pittsburgh, PA

Graduate Research Fellowship

2023

University of Waterloo

Waterloo, Ontario

Waterloo Statistics Student Conference Presentation Award

2022

Carnegie Mellon University Department of Statistics and Data Science

Pittsburgh, PA

Teaching Assistant of the Year

2021

Adobe Research

Pittsburgh, PA

PhD Research Gift

2020

University of Waterloo

Waterloo, Ontario

David Johnston International Experience Award

2018

The Natural Sciences and Engineering Research Council of Canada

Waterloo, Ontario

NSERC Undergraduate Student Research Award

2017

University of Waterloo

Waterloo, Ontario

President's Research Award

2016–17

University of Waterloo

Waterloo, Ontario

University of Waterloo President's Scholarship of Distinction

2014

Presentations

Short courses.....

MBZUAI

A brief introduction to game-theoretic, safe, anytime-valid inference

Mini course consisting of 4 lectures

Abu Dhabi, UAE

2025

ERC OCEAN retreat

A brief introduction to game-theoretic, safe, anytime-valid inference

Mini course consisting of 3 lectures

Venice, Italy

2024

Department seminars.....

Department of Statistics at the London School of Economics

Log-optimality and multi-armed sequential hypothesis testing

London, UK

2026

Department of Statistics and Actuarial Science at the University of Waterloo

Log-optimality and multi-armed sequential hypothesis testing

Waterloo, Ontario

2026

Department of Statistics at University of California, Davis

Log-optimality and multi-armed sequential hypothesis testing

Davis, CA

2026

Department of Statistics at Iowa State University

Anytime-valid inference and uniform central limit theory

Ames, IA

2026

Division of Biostatistics at the University of California, Berkeley

Anytime-valid off-policy inference for contextual bandits

Berkeley, CA

2025

Department of Statistics at Stanford University

Anytime-valid inference and uniform central limit theory

Stanford, CA

2025

Statistics Seminar at the University of Copenhagen

Anytime-valid off-policy inference for contextual bandits

Copenhagen, Denmark

2023

Invited talks at conferences / workshops / research groups.....

Institute for Operations Research and the Management Sciences (INFORMS)

Multi-armed log-optimality and sequential testing

San Francisco, CA

2026

Joint Statistical Meetings (JSM)

Asymptotic anytime-valid inference

Boston, MA

2026

Adaptive Experimentation Workshop at University College London (Gatsby)

Log-optimality and multi-armed sequential hypothesis testing

London, UK

2026

International Workshop on Sequential Methodologies

Asymptotic anytime-valid statistical inference

Washington, D.C.

2026

Recipient of the IMS New Researcher Travel Award

Stanford Causal Inference Reading Group (host: Vasilis Syrgkanis)

On nonasymptotic confidence intervals for treatment effects in randomized experiments

Stanford, CA

2026

National University of Singapore Young Mathematical Scientists Forum

Log-optimality of e-processes and sequential tests

Singapore

2025

Stanford Data Driven Seminar (host: Stefan Wager)

Log-optimality of e-processes and sequential tests

Stanford, CA

2025

MBZUAI-Berkeley Workshop

Anytime-valid off-policy inference for contextual bandits

Abu Dhabi, UAE

2025

Inria/Sierra Seminar

Anytime-valid inference and uniform central limit theory

Paris, France

2025

International Seminar on Selective Inference <i>$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X</i>	Virtual 2024
CLIMB Workshop <i>Election audits via anytime-valid inference</i>	Berkeley, CA 2024
Workshop on Game-Theoretic Statistical Inference <i>$\mathcal{P}$-uniform anytime-valid inference and conditional independence testing without Model-X</i>	Oberwolfach, Germany 2024
Fienberg Student Research Workshop at Carnegie Mellon University <i>Election audits via anytime-valid inference</i>	Pittsburgh, PA 2024
Joint Statistical Meetings <i>Anytime-valid off-policy inference for contextual bandits</i>	Toronto, Ontario 2023
International Conference on Machine Learning <i>Extensions of randomized response for private confidence sets</i>	Honolulu, HI 2023
Centrum Wiskunde & Informatica (hosts: Peter Grünwald & Wouter Koolen) <i>Anytime-valid off-policy inference for contextual bandits</i>	Amsterdam, Netherlands 2023
University of Toronto / The Vector Institute (host: Dan Roy) <i>Anytime-valid off-policy inference for contextual bandits</i>	Toronto, Ontario 2023
Copenhagen Causality Lab, University of Copenhagen <i>Asymptotic confidence sequences for anytime-valid causal inference</i>	Virtual 2023
Conference on Digital Experimentation (CODE@MIT) <i>Asymptotic confidence sequences for anytime-valid causal inference</i>	Cambridge, MA 2022
Microsoft Research Reinforcement Learning Discussion Group <i>Anytime-valid contextual bandit inference</i>	Virtual 2022
CMS, California Institute of Technology (host: Houman Owhadi) <i>A brief introduction to safe, anytime-valid inference (SAVI)</i>	Virtual 2022
Microsoft Research <i>A brief introduction to safe, anytime-valid inference (SAVI)</i>	Virtual 2022
Safe, Anytime-Valid Inference (SAVI) Workshop <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Eindhoven, Netherlands 2022
Carnegie Mellon University Computer Science Theory Lunch <i>Estimating means of bounded random variables by betting</i>	Pittsburgh, PA 2021
International Seminar on Distribution-Free Statistics <i>Estimating means of bounded random variables by betting</i>	Virtual 2021
E-Vote-ID: The International Conference for Electronic Voting <i>RiLACS: Risk-limiting audits via confidence sequences</i>	Virtual 2021
Spotify Experimentation Platform Team <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
Vinted Science and Analytics Meetup <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
Other talks	
International Conference on Statistics and Data Science (ICSIDS) <i>Concentration inequalities for strong laws of large numbers</i>	Seville, Spain 2025
Statistical Society of Canada meeting <i>Distribution-uniform strong laws of large numbers</i>	St. John's, NL 2024

Recipient of the Probability Section's Student Presentation Award

International Conference on Statistics and Data Science (ICSDS) <i>Distribution-uniform anytime-valid inference</i>	Lisbon, Portugal 2023
Waterloo Student Conference in Statistics, Actuarial Science, and Finance <i>Estimating means of bounded random variables by betting</i> Recipient of the Student Presentation Award	Waterloo, Ontario 2022
TPDP: Theory and Practice of Differential Privacy Workshop <i>Locally private nonparametric confidence intervals and sequences</i>	Baltimore, MD 2022
Statistical Society of Canada (SSC) Annual Meeting <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Virtual 2022
NeurIPS Workshop on Causal Inference Challenges in Sequential Decision Making <i>Time-uniform central limit theory and anytime-valid causal inference</i>	Virtual 2021
Joint Statistical Meetings (JSM) <i>Doubly robust confidence sequences for sequential causal inference</i>	Virtual 2021
Joint Statistical Meetings (JSM) <i>Confidence sequences for sampling without replacement</i>	Virtual 2020
Statistical Society of Canada (SSC) Annual Meeting <i>Multi-state models for chronic kidney disease prevalence projections in Ontario</i>	St. Catherines, Ontario 2016